AN ERROR DETECTION METHOD FOR UNWRAPPED INSAR PHASES BASED ON THE UNET ARCHITECTURE

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***Abstract*—Phase unwrapping (PU) is critical in the interferometric synthetic aperture radar (InSAR) technique. Since PU is an ill-posed problem, unavoidable errors in wrapped phases exist and can adversely affect the subsequential applications with the phases. Then, error detection and correction become essential. A deep learning-based approach, the PUED-UNet network, is proposed to detect errors in unwrapped phases. Analyses show that the network performs well in error detection, both qualitatively and quantitatively. Thus, a novel method for evaluating errors has been developed, which should establish a solid foundation for further error correction strategies.**

Keywords—Deep learning; Error detection in unwrapped phases; PUED-UNet; Wrapped phases in InSAR data.

I. INTRODUCTION

The interferometric synthetic aperture radar (InSAR) technique inherits the excellent characteristics of SAR in high-resolution and day-and-night and weather-independent capabilities in earth observation. With SAR data acquired from slightly different positions, the technique can accurately measure geophysical parameters using SAR data's phase information. However, the technique only obtains the phase module 2π, i.e., a wrapped phase. The phase should be unwrapped before the related applications. Therefore, phase unwrapping (PU) is indispensable for determining the technique's mapping accuracy and application potential [1]. The PU process can be expressed as (1), where $\psi(i, j)$ is the desired absolute or unwrapped phase of the pixel located at ith row and jth column of the image. $\varphi(i, j)$ is the wrapped phase, and $k(i, j)$ is an ambiguity number.

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$$\psi(i,j) = \varphi(i,j) + 2k(i,j)\pi, \quad k(i,j) \in \text{integer} \tag{1}$$

From (1), the process is an ill-posed problem because it has two unknowns, $\psi(i, j)$ and $k(i, j)$. The phase ambiguity cannot be eliminated solely through (1), necessitating the incorporation of additional information to perform the PU effectively.

Existing single baseline PU methods address this issue using the phase continuity assumption to introduce supplementary information [2]. The assumption requires that the absolute phase gradient is less than π. However, phase aliasing, often caused by noisy or abrupt topographic changes, can invalidate the assumption, producing errors after the PU methods.

In mitigating the problem, multiple single baseline PU methods have been developed, and they can be broadly grouped into path-following and optimization-based [1]. The former aims to identify residues and layout branch cuts to find a path-independent PU solution. The latter formulates the PU problem as a global minimization problem subject to integer constraints. Then, deep learning-based PU algorithms, considered the third category, have recently been developed, leveraging empirical information to accomplish phase recovery. Based on the processing steps, deep learning-based PU algorithms can be divided into one-step and two-step methods.

For optimization-based PU algorithms, the L^p-norm is often employed as a criterion to minimize the residuals, with weights $w(i, j)$ applied to the difference between unknown and estimated unwrapped gradients [1]. It can be written as

$$\begin{aligned} \min \quad & \sum_{(i,j)} w(i,j)\,|\,t(i,j)\,|^p \\ \text{s.t.} \quad & \Delta k(i,j) - \overline{\Delta} k(i,j) = t(i,j) \\ & \Delta k(i,j), \overline{\Delta} k(i,j) \in \text{integer} \end{aligned} \tag{2}$$

Here, $\Delta k(i, j)$ and $\overline{\Delta} k(i, j)$ are gradients of the integer ambiguity numbers for the unwrapped phase and the

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estimated unwrapped phase, in the row and column directions, respectively. $t(i, j)$ is an auxiliary variable. Different values of p correspond to different PU algorithms. For instance, when $p = 1$, (2) is the minimum cost flow (MCF) method [3]. Moreover, when $p = 0$, (2) becomes a weak *NP*-Hard problem, making it theoretically intractable [4]. In such cases, approximate solutions should be pursued rather than optimal solutions. For $p > 2$, PU errors tend to propagate throughout the solution, rendering the PU results unusable [1][3].

The fundamental assumption underlying single baseline PU algorithms makes the InSAR technique particularly vulnerable to decorrelation factors, leading to inevitable unwrapped phase errors [5]. Moreover, a significant challenge is that these errors cannot be effectively assessed without relying on external data, typically derived from existing products or acquired through other techniques [4]. Unfortunately, the external data cannot be readily available due to the rapidly growing InSAR data from advanced SAR satellites. Thus, the PU error detection and correction algorithms have received significant attention and have undergone rapid development [5][6]. The algorithms, relying solely on InSAR data, are essential for advancing the development and application of the InSAR technique.

Before the error correction, the error detection is critical [5]. Existing studies typically identify errors based on temporal or spatial information. The availability of the information can be challenged sometimes. This study introduces a UNet-based error detection method for the PU, termed PUED-UNet, considering deep learning's powerful information extraction capabilities. The PUED-UNet offers two capabilities: i) to detect errors in existing PU algorithms, enabling the evaluation of PU results without external data, and ii) to generate error distribution data intrinsically from InSAR data, leveraged for further optimization of unwrapping results. In summary, it is anticipated that the PUED-UNet can achieve end-to-end error detection from unwrapped results. It can enhance the detection accuracy and significantly simplify the traditional multi-step processing workflow [5]. This study should serve as a preliminary but fundamental step forward for error correction.

II. Dataset and Method

A. Dataset generation

The InSAR-DLPU dataset [7] was used to generate the required training and testing dataset. It comprises 31,000 pairs of wrapped and corresponding absolute phases from comprehensive terrain data across China. The size of each pair is 256×256. The dataset used by PUED-UNet was constructed by unwrapping the wrapped phase using the MCF method [3] to obtain the unwrapped phase dataset. The unwrapped results were then subtracted from the absolute phase dataset to generate corresponding error labels. The MCF PU results and error labels make up the final dataset. In addition, the mathematical formulation of MCF is achieved by setting $p = 1$ in (2). Fig. 1 shows one example of the training data and its corresponding label. Fig. 1 (a) and (b) are the wrapped and absolute phases from the InSAR-DLPU dataset, respectively. Fig. 1 (c) shows the PU results from the MCF method, and Fig. 1 (d) indicates the error labels derived by subtracting the unwrapped phase (c) from the absolute phase (b).

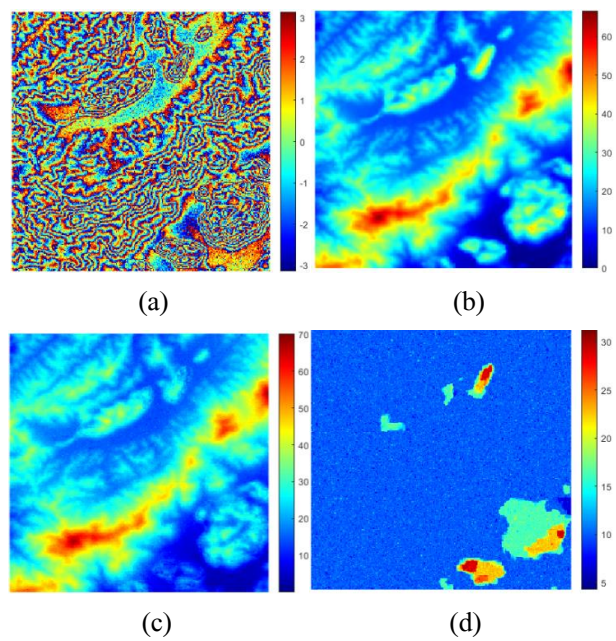

Fig. 1. One example of the training dataset. (a) The training wrapped phase from the InSAR-DLPU dataset; (b) The training absolute phase of (a); (c) The MCF PU result of (a); and (d) Error labels corresponding to (c) by subtracting (c) from (b).

B. Architecture of the PUED-UNet and training detail

The PUED-UNet, as shown in Fig. 2, adopts a multi-scale UNet-like architecture for the feature extraction and detection tasks. The network has two main components: a contracting path and an expansive one designed to ensure effective multi-scale feature extraction and fusion. The contracting path progressively reduces spatial resolution through max-pooling layers and incorporates residual blocks for efficient feature learning. The blocks enable deeper feature extraction while maintaining gradient flow and preventing degradation. The expansive path restores the spatial resolution with the deconvolution layers and skips connections to concatenate features from the

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contracting path. This design ensures that low-level spatial details and high-level semantic information are preserved and fused effectively.

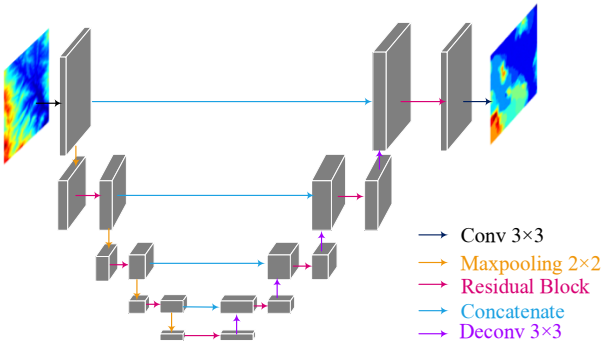

Fig. 2. The PUED-UNet. It is a U-Net-like structure enhanced with residual blocks to improve feature learning and network performance.

By leveraging this multi-scale structure, the PUED-UNet facilitates accurate detections while maintaining spatial and contextual information. Specifically, the input to the network consists of unwrapped phase results from the MCF, and the labels represent the corresponding errors after unwrapping. The network output is the error map, enabling accurate error estimation and providing a foundation for improving PU accuracy. (As the PUED-UNet is trained using PU results and error labels, it can also detect errors in unwrapping results generated by other algorithms rather than the MCF per se.)

In this study, the dataset was divided into a training set (30,000 pairs) and a test set (1,000 pairs). The network was trained for 50 epochs using the Adam optimizer with a 1×10^{-4} learning rate. All training and testing processes were performed on a computer with an Intel i7-12700KF CPU and an NVIDIA GeForce RTX 3090 GPU with 24 GB of memory.

III. Results and Discussions

After the training, the PUED-UNet was applied to the test dataset. Fig. 3 shows, as an example, detection results. The first column represents the input MCF unwrapped results, and the second column shows the corresponding unwrapping errors (ground truth). The third column displays detected errors, and the fourth column presents the differences between the labels and detected errors. The corresponding RMSE value is at the bottom-right corner (column 4). The average RMSE value for all 1,000 test samples is 0.8484, indicating the network's satisfactory error-detection capability.

Considering the perspective of PU principles, the L^1-norm can effectively constrain errors to a limited number of points or subregions in areas with gentle terrain or sparse interferometric fringes. However, in regions with complex terrain, the phase continuity assumption leads to increasing invalidation of the irrotationality of the phase gradient field. This results in the propagation of unwrapping errors along the integration path, causing severe phase errors after a PU. The residual maps reflect the errors as increasingly mottled and fragmented patterns.

To support the arguments, we show Fig. 3 based on the complexity of the terrain or ground truth by rows, ranging from simple to complex. The residual maps (2nd column) exhibit uniform color regions for the MCF unwrapped results in better detection accuracy, as demonstrated in Fig. 3(d) of row one). It contains only isolated pixel-level errors or small blocks. In contrast, for unwrapping results with severe errors, their residual maps exhibit increasingly pronounced block-like artifacts and texture information (Fig. 3(p) of row four). The PU results become less accurate as the error patterns become complicated. Quantitatively, the corresponding RMSE values (near the lower-right corners of the fourth column) increase by rows as the complexity of the residual maps grows. Overall, the detection results capture the characteristics and distribution of errors, enabling the identification of regions where the MCF method exhibits significant unwrapped phase errors. In short, the PUED-UNet effectively detects errors after a PU method and lays a solid foundation for subsequent error correction efforts.

IV. Conclusion

A PUED-UNet network was studied to detect PU phase errors after using the PU method. It detects errors numerically and spatially (assessed by error distributions). The analyses demonstrate that the network performs qualitative and quantitative error estimations effectively. Thus, a solid foundation for subsequent error correction has been provided. Specifically, the error distribution data generated by the deep learning model enables more accurate refinement of the existing InAR unwrapping methods through optimization procedures. Additionally, the study confirms that evaluating unwrapping results with existing and empirical datasets is possible for the InSAR technique. Using deep learning networks to detect and even predict unwrapped phase errors and their distributions after a PU method is also feasible. Thus, the study holds significant promise for improving the accuracy of InSAR applications.

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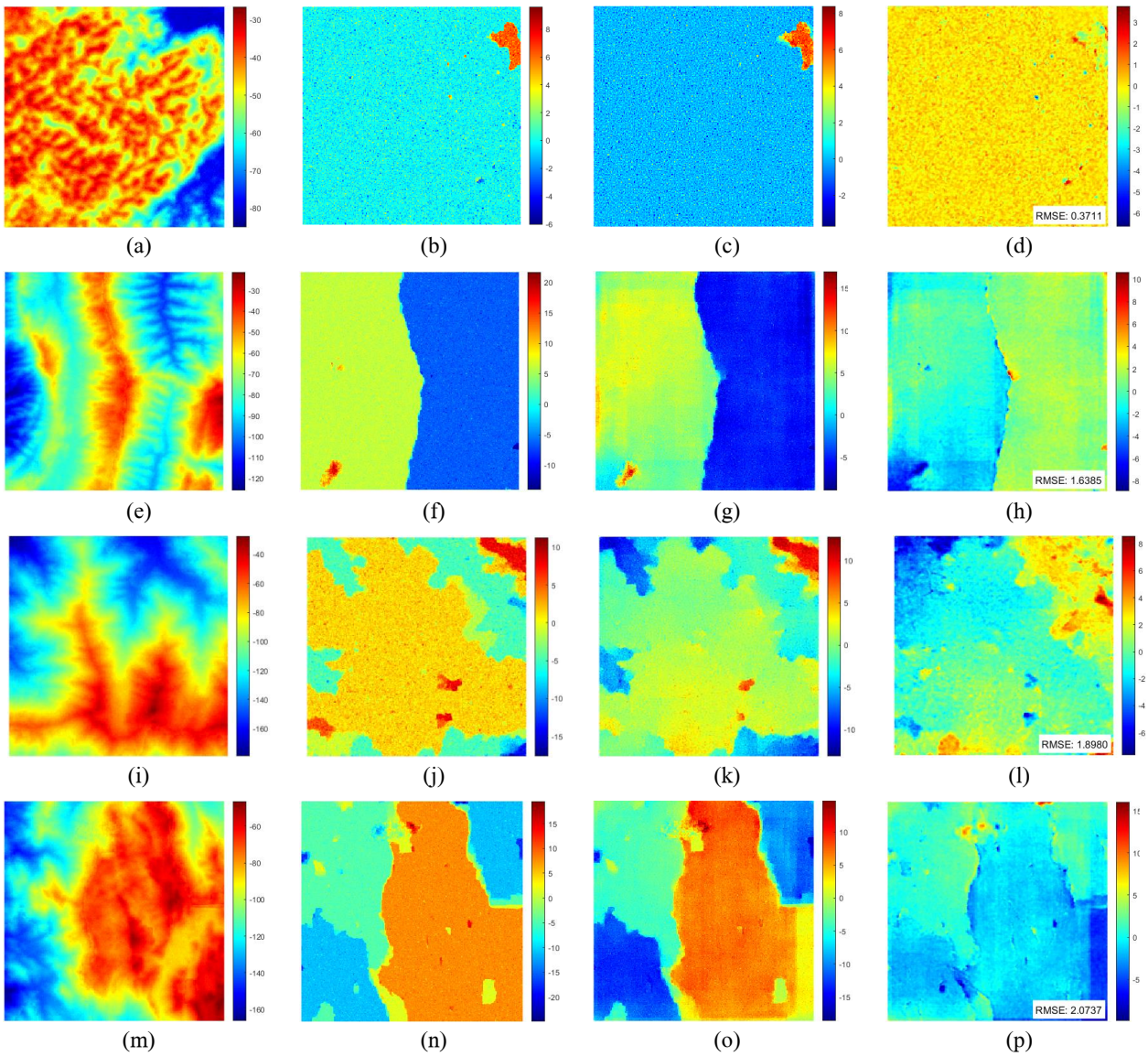

Fig. 3. PU error detection results of PUEP-UNet. Column one: The input MCF unwrapped results from the test dataset; column two: the corresponding ground truth; column three: error detection results by the PUEP-UNet; and column four: difference between the second and third columns, by rows. The complixity of the ground truh (column two) increase from rows 1 to 4.

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